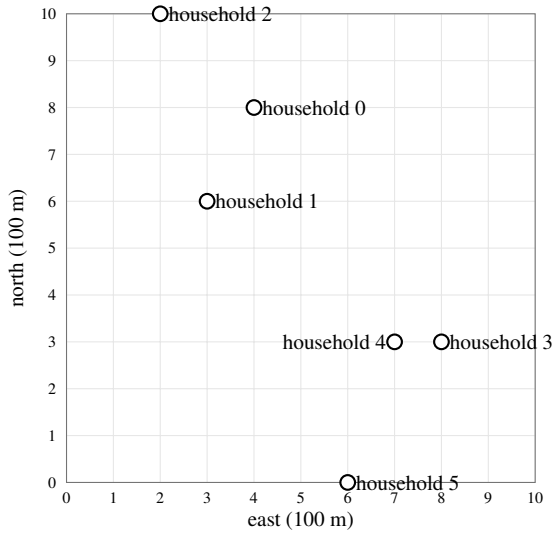
Where to park two vans, by hand

Lecture 10 | Six synthetic households of survey A, $K = 2$ vans, units of 100 m



household	east	north
0	4	8
1	3	6
2	2	10
3	8	3
4	7	3
5	6	0

The squared distance from x to c is $(x[0] - c[0])^2 + (x[1] - c[1])^2$.
The within-cluster sum of squares W adds, over all households, the squared distance to the center of the household's own cluster.

1. The W of a split. Clusters $\{0, 1, 2\}$ and $\{3, 4, 5\}$ with centers $(3, 8)$ and $(7, 2)$. The six squared distances: _____, _____, _____, _____, _____, _____. $W =$ _____

2. Round 1. Starting centers $\mu_0 = (4, 8)$ (household 0) and $\mu_1 = (3, 6)$ (household 1). Give each household the label of the nearer center.

household	point	to $\mu_0 = (4, 8)$	to $\mu_1 = (3, 6)$	label
0	$(4, 8)$	_____	_____	____
1	$(3, 6)$	_____	_____	____
2	$(2, 10)$	_____	_____	____
3	$(8, 3)$	_____	_____	____
4	$(7, 3)$	_____	_____	____
5	$(6, 0)$	_____	_____	____

Learning step: each new center μ_k is the centroid (mean) of cluster C_k. $\mu_0 =$ _____ $\mu_1 =$ _____

3. Round 2. Use your new centers.

household	point	to μ_0 = _____	to μ_1 = _____	label
0	$(4, 8)$	_____	_____	____
1	$(3, 6)$	_____	_____	____
2	$(2, 10)$	_____	_____	____
3	$(8, 3)$	_____	_____	____
4	$(7, 3)$	_____	_____	____
5	$(6, 0)$	_____	_____	____

Which household changed its label? _____ New centers: μ_0 = _____ μ_1 = _____

4. Survey B. Six other households from the same city. k-means stops with the clusters $\{(1, 7),\ (3, 9)\}$ and $\{(9, 4),\ (6, 4),\ (7, 0),\ (10, 4)\}$, which is the best split of these six.

Centers: _____ and _____

Survey A's run also returns the best split of its own six households. Why do the two surveys give different centers?

_____

_____

5. Exit check (when prompted). A third survey records six other households from the same city, and k-means with $K = 2$ on them gives the centers $(4, 7)$ and $(6, 2)$. The city's centers, from its best split, are $(3, 7)$ and $(7, 3)$. Give two different reasons why these centers can differ from the city's.

(a) _____

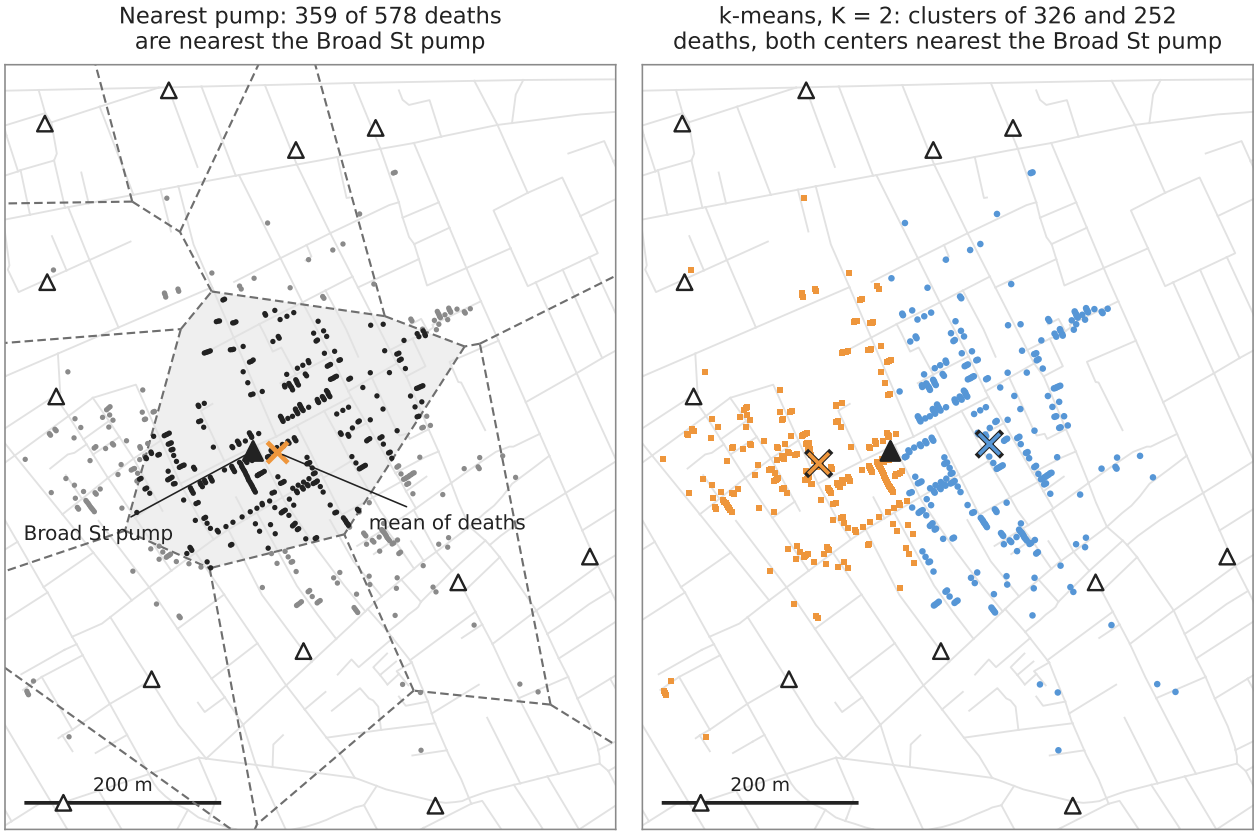
_____

(b) _____

_____

Snow's map, 1854

578 cholera deaths and the 13 public pumps in Soho, London (digitized)



Left: each death has the label of its nearest pump by straight-line distance, and the Broad Street region is shaded. Right: k-means with $K = 2$ on the deaths alone, with the pumps ignored.

6. Think, then discuss (when prompted). The orange $\times$ on the left is the mean of all 578 deaths. It lies about 25 m from the Broad Street pump.

(a) Which part of k-means does the left panel show? Why is there no learning step?

(b) Before reading the position of the mean as evidence that the Broad Street pump was the source, what would you compare it with?

(c) What does the right panel say about reading a cluster as a source?